

Homework-4

1. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \dots, \sqrt{p}, \dots)$, p a prime. Prove that $E/\mathbb{Q}$ is not a finite extension. Let $L \subseteq E$ be a finite subextension of $\mathbb{Q}$. Prove that $[L:\mathbb{Q}] = 2^r$ for some $r \geq 1$.
2. Let R be an integral domain containing a field k . Assume, with the natural scalar action of k on R , $\dim_k R < \infty$. Prove that R is a field.
3. Let K_1, K_2 be subfields of a field K , both containing a field k . Assume $\gcd([K_1:k], [K_2:k]) = 1$. Prove that $[K_1 K_2:k] = [K_1:k][K_2:k]$.
4. Let k be a field and $k(x)$ be the function field in 1-variable. Let $k \subsetneq L \subsetneq k(x)$ be a subextension. Prove that $k(x)$ is algebraic over L .
5. Let $f_1(x), \dots, f_m(x) \in k[x]$. Prove that $\exists$ a finite extension L of k where all the f_i 's have a root. Hence or otherwise prove that $\exists$ a field E/k , $[E:k] < \infty$, splitting all the f_i 's simultaneously.
6. Call a field extension K/k simple if $K = k(\alpha)$ for some $\alpha \in K$. Suppose K is a finite extⁿ of k and that the set $\{L \subseteq K \mid L \text{ a subextⁿ of } k\}$ is totally ordered with $\subseteq$. Prove K/k is simple.

7. Prove that $\overline{\mathbb{Q}} := \{\alpha \in \mathbb{C} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}$ is algebraically closed (i.e. $\alpha \in \overline{\mathbb{Q}} \text{ alg} \Rightarrow \alpha \in \overline{\mathbb{Q}}$).

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8. Prove that a field extⁿ K/k is algebraic if and only if every ring R s.t. $k \subseteq R \subseteq K$ is a field.
9. Prove that $\overline{\mathbb{Q}}$ is countable. This proves $\mathbb{R}$ is a transcendental extⁿ of $\mathbb{Q}$; same for $\mathbb{C}/\mathbb{Q}$.
10. Is it possible to have a finite field $\mathbb{F}$ that is algebraically closed? Explain.
11. Let $K = k(\alpha)$ be a finite extⁿ with $[k(\alpha):k]$ odd. Prove $K = k(\alpha^2)$.
12. For every $n \in \mathbb{N}$, $\exists$ an extⁿ $L/\mathbb{Q}$ with $[L:\mathbb{Q}] = n$.
13. Let $K = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$ with $\alpha_i^2 \in \mathbb{Q}$, $1 \leq i \leq n$. Can $2^{1/3} \in K$? Explain.
14. Let k be a field, $\text{Char}(k) \neq 2$. Let $\alpha, \beta \in k \setminus \{0\}$, but both $\alpha, \beta \notin k^2$ ($k^2 = \{\alpha^2 \mid \alpha \in k\}$). Prove that $k(\sqrt{\alpha}, \sqrt{\beta})$ is of degree 4 over k if $\alpha\beta$ is not a square in k and is of degree 2 otherwise.
15. Let k be a field; α algebraic over k . Let $L_\alpha: k(\alpha) \rightarrow k(\alpha)$ be the map $L_\alpha(x) = \alpha x$, $\forall x \in k(\alpha)$. Clearly L_α is k -linear. Prove that $\det(xI - L_\alpha) = \text{Irr}(\alpha, k)$.